the algorithm doesn't make use of the key directly but first obtains the 128-bit key.

Now, some of you will be wondering how on earth we're going to encrypt a 32-bit block-size plaintext with a 128-bit key size. Also, if we're using XOR, how do you XOR a 32-bit block with a 128-bit block? That's what we'll show you in the next step.

NOTE: There are better ways to get a 128-bit key, given another key is known. One of the best ways is to use hashing techniques. A good example would be the MD5 algorithm. This algorithm takes in a variable sequence of inputs.

Step 4

Since the plaintext block size is 32 bit and the key length is 128 bit, they cannot be involved in a direct XOR operation. So what we do here is take consecutive 32-bit blocks from the plaintext and XOR them with consecutive 32-bit blocks of the key. Following this method, the key would get consumed by the fourth block of the plaintext (32 x 4 = 128). The fifth block of the plaintext would then be XORed with the first 32-bit block of the key, and so on. Hence, if the plaintext is "FDA72BE910CA8AB3C1068EE-A735715BCFF2938A31029ABD37ECA928CE7EA53C3" and the 128-key is "02468ACECE02468A8ACE0246468ACE02" (which we got in Step 2) then the XOR operation would be done with chunks in the following manner.

TABLE 3: GETTING THE ACTUAL CIPHERTEXT USING THE PLAINTEXT AND THE 128-BIT KEY

Part of plaintext	FDA72BE9	10CA8AB3	C1068EEA	735715BC	FF2938A3	1029ABD3	7ECA928C	E7EA53C3
Part of key	02468ACE	CE02468A	8ACE0246	468ACE02	02468ACE	CE02468A	8ACE0246	468ACE02
Resulting ciphertext	FFE1A127	DEC8CC39	4BC88CAC	35DDDBBE	FD6FB26D	DE2BED59	F40490CA	A1609DC1

We're going to concatenate the 32-bit blocks of result into one to get the ciphertext. Hence the resulting ciphertext would be:

FFE1A127DEC8CC394BC88CAC35DDDBBEFD6FB26DDE2BED59F-40490CAA1609DC1

Decryption

Decryption of the ciphertext is straightforward in this algorithm. Of course, the two things we're going to need before beginning the process are the original 32-bit key and the entire ciphertext. We then are going to derive the 128-bit key from the 32-bit key using the process in Step 3. Once we have